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SUPERSEDING
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MILITARY SPECIFICATION

ROLLERS, TOWED, SMOOTH DRUM, VIBRATORY, AIRMOBILE, SINGLE-DRUM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a diesel-engine-powered, towed-type, smooth-drum, vibratory roller.

1.2 Classification. Rollers shall be one of the following types, as specified (see 6.2):

Type I - 30-inch drum diameter.
Type II - 48-inch drum diameter.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-P-664	- Primer, Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.
VV-F-800	- Fuel Oil Diesel.
ZZ-B-225	- Belts, V: Industrial (Multiple Drive).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3895

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MIL-V-173	- Varnish, Moisture and Fungus-Resistant, (For Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-L-2104	- Lubricating Oil, Internal Combustion Engine Tactical Service.
MIL-R-3075	- Roller, Motorized, Road, Diesel or Gasoline-Engine-Driven, Packaging of.
MIL-F-3541	- Fittings, Lubrication.
MIL-G-3859	- Grease Guns, Hand, High-Pressure, Lever-Operated; Cartridge and Bulk Loading (14 Ounce and 21 Ounce Capacity).
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-T-52510	- Tractor, Full Tracked, Low-Speed, Diesel Engine-Driven, Air-Droppable and Helicopter Transportable.
MIL-E-52929	- Enamel, Alkyd, Camouflage, Flash Dry.

STANDARD

MILITARY

MIL-STD-130	- Identification Marking of U.S. Military Property.
MIL-STD-209	- Slinging Eyes and Attachments for Lifting and Tying Down of Military Equipment.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1474	- Noise Limits for Army Material.
MS35489	- Grommet, Rubber, Hot Oil and Coolant Resistant.
MS35643	- Gasket: Cap, Fuel Tank.
MS35644	- Filler Neck Assembly, Fuel Tank-Military Vehicles.
MS35645	- Cap, Fuel Tank-Tactical Vehicle, Fording.
MS53063	- Indicator, Air Cleaner, Intake; Restriction Mechanical Type.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

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SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The smooth drum, vibratory, airmobile, single drum, towed roller (hereinafter called "roller") shall consist of a vibrating mechanism contained within a drum, a frame, a front drawbar with towing hitch, a rear clevis, replaceable drum cleaners, diesel engine, vibrating mechanism and remote control.

3.1.1 Performance. The roller shall withstand rough operation encountered when towed over loose fill dirt, excavation areas, and rock fill areas. The roller shall be capable of performing at its rated capability when towed by crawler tractors, rubber-tired tractors or trucks (as required). The roller shall be capable of performing as specified herein after being transported by truck, trailer or as an internal or slung load from helicopter organic to airmobile military operations.

3.1.2 Dimensions and weight. The type I and type II rollers shall conform to the requirements listed in table I.

TABLE I. Dimensions, weight and dynamic force.

Requirements	Type I	Type II
Overall height, maximum	72 in.	72 in.
Overall width, maximum	68 in.	88 in.
Drum diameter	28 to 32 in.	48 to 52 in.
Drum width	54 to 60 in.	58 to 66 in.
Drum shell and head thickness, minimum	5/8 in.	3/4 in.
Empty weight, maximum	5000 lb.	10,000 lb.
Centrifugal force, minimum	10,000 lb.	18,000 lb.

3.2 First article (preproduction model). Unless otherwise specified (see 6.2), the contractor shall furnish one or more rollers for examination and testing within the time frame specified

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(see 6.2), to prove prior to starting production that his production methods and choice of design detail will produce rollers that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and unless otherwise specified herein, all examinations and tests shall be conducted by the contractor subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the preproduction examination and tests, as specified (see 6.2).

3.3 Production rollers. All rollers furnished under this specification shall be physically and mechanically identical to the first article rollers approved by the Government (see 6.3).

3.4 Material. Material shall be as specified herein. Material not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.6). Materials and special treatments utilized shall provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the item may be exposed.

3.4.1 Dissimilar metals. Dissimilar metals as defined in MIL-STD-889 shall not be used in intimate contact without suitable protection in order to prevent or minimize galvanic corrosion.

3.4.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and subcomponents, and shall make information available, upon request, to the contracting officer or his/her designated representative.

3.5 Safety and human factors.

3.5.1 Safety. All rotating or reciprocating parts, and all parts subject to high operational temperature that are of such nature or are so located as to be, or become, a hazard to operating or maintenance personnel shall be guarded or insulated to the extent necessary to eliminate the hazard. All platforms, steps, and ladder rungs shall have antiskid surfaces. All exhausts shall be directed so as not to endanger personnel or obstruct the view of the operator. Fuel tanks shall be located in a manner which will not allow spills and overflows to run onto the engine, exhaust, or electrical equipment. Electrical equipment shall be insulated, fully enclosed, or guarded. Nonfunctional sharp edges, projecting points and protruding length more than two full threads of fastening devices shall be avoided.

3.5.2 Noise level.

3.5.2.1 Exterior noise. The exterior sound level of the roller shall not exceed 88dB(A) when measured in accordance with SAE J88.

3.5.2.2 Personnel noise. The sound level at the operator's starting position for starting the roller shall be 85dB(A) or less when measured in accordance with SAE J919.

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3.6 Maintenance.

3.6.1 Maintainability. All major assemblies and installed attachments shall be accessible for maintenance, repair, and replacement with minimum removal of other major assemblies and installed attachments. Covers or plates which must be removed for component adjustment or maintenance shall be equipped with quick-disconnect fastenings. Drain outlets shall be accessible. Ease of maintenance provisions shall incorporate features insuring operating clearances for handtools and lubricating equipment, and the accessibility for service operations such as adding oil to the engine crankcase and adding oil to gear cases. Each maintenance, assembly, or disassembly operation required during testing in accordance with 4.3 shall be accomplished by no more than two men (except a hoist and operator may be used as necessary) using special tools furnished with the roller and common tools. After completion of any maintenance operation, the roller shall operate as specified herein.

3.6.2 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Common hardware and fastening devices such as bolts, nuts, screws, and washers shall be held to a minimum number of types and sizes.

3.7 Frame. The frame shall be constructed of structural steel shapes or plates. The frame shall be isolated from the drum and vibrating mechanism. The frame shall incorporate a drawbar and adjustable retractable support leg.

3.7.1 Drawbar. The drawbar shall permit pulling and pushing the roller. The roller shall be capable of being coupled for towing after being stabilized in parking position by the support leg (see 3.7). Unless otherwise specified, the drawbar shall be equipped with a lunette ring. The lunette ring shall conform to SAE J847. The drawbar lunette ring mounting shall provide a lunette ring height range from 27 to 33 inches.

3.7.2 Crawler tractor drawbar. When specified (see 6.2) the roller drawbar and lunette ring shall be compatible with a crawler tractor drawbar conforming to SAE J749a, pin size 1.75.

3.7.3 Support leg. An adjustable support leg shall be mounted on the drawbar. The support leg shall be vertically adjustable for not less than 12 inches. The support leg shall either retract or be hinged and fold to provide not less than 10 inches ground clearance. The support leg shall be arranged for positive locking into park and operational (travel) position. The support leg shall incorporate a self-alining pad which will provide a ground bearing load not greater than 20 pounds per square inch when the maximum drawbar weight is imposed. The edges of the pad shall be turned up.

3.8 Drum. The shell and heads of the drum shall be constructed of steel plates, which shall be of the thickness specified in table I. The outer surface of the shell shall be smooth. The joints between shell and heads shall be welded and the heads shall not incorporate provisions for ballasting.

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3.9 Axles and bearings. The drum shall be supported to provide independent rolling capabilities. The vibrating shaft shall extend entirely through the drum. The vibrating shaft may act as a drum axle through the use of a hollow collar connected to the drum and turning on the vibrating axle shaft. The vibrating axle and drum shall be equipped with either antifriction bearings or bronze bushings or graphite impregnated phenolic resin laminated bushings. Appropriate lubrication points shall be provided. The drum support bearings shall have adequate thrust capacity for operation under all field conditions.

3.10 Cleaners. Replaceable cleaners shall be provided over the full width of the drum and shall prevent accumulation of dirt on the drum when the roller is towed in both the forward and reverse direction. The cleaners shall be adjustable to provide a minimum clearance of 3/8 inch between drum and cleaners. Since this clearance will permit dirt to pass between drum and cleaners, the cleaners shall incorporate provisions to prevent accumulation of dirt above the cleaners.

3.11 Rear clevis(s). A rear clevis shall be provided and positioned on the longitudinal centerline of the roller. Unless otherwise specified the clevis shall be of such size to accommodate a lunette as specified in 3.7.1.

3.12 Power unit. The roller shall be powered by a commercial diesel engine having horsepower, torque, and speed characteristics to satisfactorily meet all performance requirements specified herein. The engine shall be equipped with a 12 volt or 24 volt electrical cranking system. The electric cranking system shall include but not be limited to the following: an electric starter, alternator, voltage regulator, battery and all other accessories required to complete the cranking system. The engine shall be mounted as to prevent damage to the engine and attached engine external accessories by load sling cables, when the roller is being transported as a slung load from helicopters specified. The engine shall be equipped with an instrument panel. The panel shall be located in a protected enclosure and shall be isolated so as to dampen the vibration effects. The engine shall be provided with the following accessories:

a. Ammeter or voltmeter.
b. Oil pressure indicator.
c. Oil filter.
d. Air cleaner, dry type.
e. Throttle control.
f. Coolant temperature gage (as applicable).
g. High temperature cut off switch.
h. Low oil pressure cut off switch.
i. Hourmeter-tachometer (combination unit).

The engine shall be capable of operating continuously at rated load in an ambient temperature of 120°F without exceeding the maximum allowable best temperatures established by the engine manufacturer. Certification to this requirement is acceptable (see 4.5.2.3).

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3.12.1 Fuel tank. A fuel tank conforming to the contractor's commercial standards, except as specified herein, shall be sufficient for 8 hours of operation at the maximum fuel rating of the engine. The tank shall be provided with fuel tank cap gasket, filler neck, and fuel tank cap conforming to MS35643, MS35644, and MS35645 respectively. The tank shall be stenciled: "FILL WITH DIESEL FUEL ONLY". The tank shall be provided with a fuel level indicator and with flexible fuel lines to the engine. Provisions shall be made to drain the fuel into a hand held container.

3.12.2 Engine fuel. The diesel engine furnished shall operate on fuel oil conforming to VV-F-800, grade DF2.

3.12.3 Engine lubrication. The engine lubricating oil shall be in accordance with MIL-L-2104.

3.12.3.1 Engine oil sampling valve. An oil sampling valve shall be provided on the engine. The valve shall be manually operated and shall automatically close after release. It shall be made of material resistant to corrosion such that it will not contaminate the sample. The discharge port of the valve shall be covered with a captive chained cap conforming to SAE J514. The valve shall be located in such a way as to insure that personnel will not be exposed to danger when taking oil samples with the engine running. The location of the oil tap shall be such that when samples are taken, it shall be a true representation of oil that is flowing while the engine is running.

3.12.4 Clutch. When a gear or belt driven vibrating mechanism is provided a hand-operated, over-center, engagement-type clutch shall be furnished. The clutch shall have a load capacity rating of not less than the maximum energy.

3.12.5 Remote controls. When specified (see 6.2) a remote control system shall be provided. The remote control system shall enable the operator to start, stop and control engine speed, engage and disengage the drum vibrator from the operators position of a crawler tractor conforming to MIL-T-52510. Provisions shall be provided to secure the controls at the operator's station of the crawler tractor during operation. Provisions shall be provided on the roller for storage of the remote control system.

3.12.6 Battery, battery box.

3.12.6.1 Battery(s). The battery(s) shall be heavy duty automotive type and shall be mounted in a battery box as specified in 3.12.6.2. The battery(s) shall furnish the power for all electrical demands. The battery(s) shall be shipped charged, dry and sealed to prevent entry of moisture during shipment and storage.

3.12.6.2 Battery box. The battery box shall be made of steel having a minimum thickness of 0.0747 inch. The battery box shall have a lid with an overlapping lip to prevent entry of rain and dirt. The lid shall be provided with not less than two overcenter locking or spring tensioned

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fasteners to secure the lid in the closed position. The battery box shall be equipped with built-in restraints so constructed that pressure will be applied to the other battery case only. The battery box shall be mounted securely to the roller and shall be isolated to dampen the vibration of the roller from the batteries and battery box. The battery cable openings shall be provided with grommets conforming to MS35489, size as required. The battery cables and grommets shall be secured to the battery box to prevent abrasive cutting by the battery box edge due to vibration. The battery box and terminals shall be coated with acid resisting compounds.

3.12.7 Air cleaner indicator. The dry-type air cleaner shall be provided with an air flow restriction indicator conforming to MS53063.

3.13 Vibrating mechanism. A vibrating mechanism shall be provided and shall develop the centrifugal force specified in table I, within the frequency range of not less than 1100 vibrations per minute and not exceeding 2400 vpm. The frequency of the vibrating mechanism shall be variable by not less than 3000 vpm within the frequency range specified herein. The vibrations shall be developed by an eccentric shaft, by eccentric weights, or by rotating balls. Gear drive or V-belt drive or hydraulic drive shall be furnished for the vibrating mechanism.

3.13.1 Gear drive. The gear drive shall include all the required gear-boxes, flexible couplings, universal drive shafts and all other necessary mechanical hardware required to provide a gear drive power train having the mechanical integrity to provide optimum operational characteristics and capabilities to operate the roller as specified in 3.1.1.

3.13.2 V-belt drive. V-belt drives shall be in accordance with ZZ-B-225. All belt drives shall be matched multiple V-belt assemblies. Provisions shall be provided to adjust belt tension. Applying tension by use of an idler will be permitted providing the idler is used on the slack side of the belt.

3.13.3 Drive lines. All drive lines shall be installed in such a manner so as to prevent damage to drive line components by load slings during transportation by helicopters organic to military airmobile operations employing slung load transportation methods.

3.13.4 Hydraulic drive. The hydraulic drive shall include the required hydraulic pump(s), motor(s), lines, control valve(s), reservoir, filter(s) and all other necessary hardwear required to provide a hydraulic drive having the operational characteristics and capabilities to operate the roller as specified in 3.1.1.

3.14. Identification marking. The roller shall be identified in accordance with MIL-STD-130. The marking shall be applied to the roller on plates conforming to MIL-P-514, Type I, Style 1, Composition C, Grade A, Class 1 material. The plates shall be attached by screws, bolts, or rivets in a conspicuous location.

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3.15 Shipping data plate. Shipping data plates shall conform to MIL-P-514, Type III, Composition C, of Type I, Grade A, Class 1 material and in addition shall show the silhouette of the roller in the transport position indicating the center of gravity and the location and capacity of the lifting and tiedown attachments. The plates shall be attached by rivets, screws, or bolts, in conspicuous protected locations.

3.16 Instruction plate. An instruction plate, conforming to MIL-P-514, Type III, Composition C, of Type I, Grade A, Class 1 material shall contain clear and concise instructions for operation of all controls and for engine start. The plate shall be permanently affixed to the roller in a conspicuous place.

3.17 Electromagnetic interference. Unless otherwise specified (see 6.2). The electromagnetic interference emission and susceptibility characteristics of the roller shall conform to SAE J551, devices category.

3.18 Lubrication. All surfaces that require lubrication shall be provided with an accessible means for lubricating.

3.18.1 Lubricants. The roller shall operate as specified herein when lubricated with standard military lubricants.

3.18.2 Lubricating fittings. Lubrication fittings shall conform to MIL-F-3541, Type I, II, or III. Fittings shall be located in a protected position and shall be accessible to a grease gun conforming to MIL-G-3859. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to provide accessibility.

3.18.3 Pressure-release device. An automatic pressure-release device shall be provided where the use of pressure-lubricating equipment will damage grease seals or other parts.

3.18.4 Filling, draining, and checking provisions. Enclosures such as gearcases that contain a reservoir of lubricants shall be equipped with dipsticks, check plugs not less than 1/4-inch pipe size, or sight gages to determine the level of the lubricant. Each enclosure shall be equipped for filling the enclosure with lubricant and for draining. The drain shall be fitted with a magnetic drainplug. The drain outlet shall be located so that removal of the plug will result in complete drainage of the lubricant from the enclosure. Drainage shall be accomplished when the roller is in its normal operating position. Integral tubes or thoughts may be used to convey the lubricant from the drain outlet to a hand held container.

3.18.5 Grease lubrication. All grease lubrication including lubrication of sealed bearings, shall be with greases conforming to MIL-G-10924 or MIL-G-23827 as applicable. The roller shall be assembled, run-in, tested, and delivered with these greases. The bearings shall be cleaned before lubrication, because military lubricants may not be compatible with other greases. A tag shall be attached in a conspicuous place to indicate which military grease has been used.

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3.18.6 Oil lubrication. Unless otherwise specified herein, all components shall be filled to the operating level with military oils designed for use in the temperature range as specified herein. A tag shall be attached in a conspicuous place to indicate the temperature range and grade of lubricant used. The high side bearings shall not be starved when the roller is operated on a 10 degree slope determined around the longitudinal axis of the roller.

3.19 Fungus and moisture resistance. Unless otherwise specified (see 6.2), the electrical circuitry, including all components and connections except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, Composition I, with 1 percent copper 8-quinolinolate (by weight) or 7 percent salicynanilide based on the nonvolatile content of varnish:

a. Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
b. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a minimum dry-film thickness of 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.20 Transportability.

3.20.1 Lifting attachments. Permanently affixed attachments that enable the roller to be lifted in its normal travel positions shall be provided. Each attachment shall meet the requirements specified in MIL-STD-209. Lifting attachments may also be used as tiedown attachments when such attachments meet the requirements specified in 3.20.2.

3.20.2 Tiedown attachments. Permanently affixed attachments that permit fastening the roller to the floor or deck of a transportation medium shall be provided. The tiedown attachments shall meet the requirements specified in MIL-STD-209. Tiedown attachments may also be used as lifting attachments when such attachments meet the requirements specified in 3.20.1.

3.21 Treatment and painting. The portions of the roller normally painted shall be cleaned, treated and painted in accordance with manufacturer's standard practice provided the anti-corrosive primer used is at least equivalent to TT-P-644 primer described in MIL-T-704 for the specified furnishing coat. Unless otherwise specified (see 6.2) the finishing coat shall be lusterless forest green in accordance with MIL-E-52929, Type II.

3.22 Stenciling. The gross weight of each roller shall be stenciled on each side of the roller. "Lift" and "Tiedown", as applicable shall be stenciled adjacent to their respective tiedown and

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lifting attachment. All stenciling shall be in block-type letters not less than 1 inch high and shall be lusterless and black in color.

3.23 Toolbox. A toolbox made of steel not less than 0.0747 inch nominal thickness shall be provided and shall be of sufficient size to hold those tools and accessories necessary to operate the roller. The toolbox shall have a hinged lid with the hinge extending the full length of one edge and shall be provided with not less than two fasteners that will keep the lid closed when the toolbox is subjected to vibration. The toolbox lid shall have an overlapping lip to prevent entry of rain and dirt. The toolbox shall be securely mounted in a protected and accessible location permitting the lid to be opened through an arc of at least 90 degrees. The lid shall remain in the fully open position by the weight of the lid or by holding devices.

3.24 Workmanship. All parts, components, and assemblies of the roller including castings, forgings, molded parts, stampings, bearings, seals, machined surfaces, and welded parts, shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux and other harmful extraneous material. All exposed edges shall be rounded or chamfered.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspection. Inspection shall be classified as follows:

a. First article preproduction inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection comparison (see 4.6).
d. Inspection of packaging (see 4.7).

4.3 First article preproduction inspection.

4.3.1 Examination. The roller shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Test. The roller shall be tested as specified in 4.5.2.1 through 4.5.2.7. Failure of any test shall be cause for rejection.

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4.4 Quality conformance inspection.

4.4.1 Examination. Each roller shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection and no rejected roller shall be accepted until all defects are corrected.

4.4.2 Test. Each roller shall be tested as specified in 4.5.2.8. Failure of the test shall be cause for rejection and no rejected roller shall be accepted until the cause and failures are corrected.

4.5 Inspection procedure.

4.5.1 Examination. The roller shall be examined for the following defects:

101. Assembly not complete or not correct.
102. Dimensions and weight not as specified.
103. Material not as specified.
104. Appearance of rust and corrosion.
105. Materials not corrosion resistant.
106. Dissimilar metals not in accordance with MIL-STD-889.
107. Material and or material finishes information not available.
108. Safety devices, guards, insulation, not as specified.
109. Maintainability not as specified.
110. Parts interchangeability not as specified.
111. Frame not as specified.
112. Drawbar not as specified.
113. Support leg not as specified.
114. Drum not as specified.
115. Drum cleaners not as specified.
116. Rear clevis(s) not as specified.
117. Power unit or power unit accessories not as specified.
118. Fuel tank not as specified.
119. Engine oil sampling valve not as specified.
120. Clutch not as specified.
121. Battery or battery box not as specified.
122. Air cleaner not as specified.
123. Vibrating mechanism not as specified.
124. Drive system not as specified.
125. Identification marking missing or not as specified.
126. Shipping data plates missing or not as specified.
127. Instruction plate missing or not as specified.
128. Lubrication devices not accessible, missing, or not installed as specified.
129. Fungus and moisture resistance not as specified.
130. Lifting and tiedown attachments not as specified.
131. Treatment and painting not as specified.
132. Stenciling not as specified.
133. Toolbox not as specified.
134. Workmanship not as specified.

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4.5.2 Tests.

4.5.2.1 Test conditions. Unless otherwise specified in a test, tests shall be performed without shelter and at the climatic conditions existing at the place of test. The roller shall operate as specified herein without maintenance other than the contractor's recommended normal scheduled maintenance as established by a maintenance schedule prepared and submitted by the contractor prior to test. Prior to testing, lubricate the roller with military lubricants as specified in 3.18. After checking and servicing the roller, connect the lunette of the roller drawbar to the drawbar or pintle of the test towing unit. When a remote control system is furnished (see 3.12.5 and 6.2) the test specified in 4.5.2.2 shall be conducted using the remote control system to control the roller operation.

4.5.2.2 Towing. Start the engine using the engine controls and engage the vibrating mechanism. Tow the roller over not less than 6 inches (type I) or 12 inches (type II) of loose dirt fill or plowed earth at a speed of not less than 1-1/2 miles per hour and not more than 3 miles per hour for 10 successive passes. One pass shall consist of not less than 100 linear yards. Repeat this test for 20 operating hours. Stop the engine once during each 2-hour operating period and restart it after a 5-minute shutdown period. During the operational test determine the VPM frequency range.

4.5.2.2.1 Failure criteria. Inability to start the engine using only the engine controls, inability to engage or disengage the vibrating mechanism, inability of the cleaners to prevent accumulation of dirt as specified in 3.10, inability to stop or restart the engine using only the engine control system, leakage of fuel or lubricants, or permanent deformation or damage shall constitute failure of this test. Inability to vary the VPM frequency range as specified (see 3.13) shall constitute failure of this test.

4.5.2.3 High temperature operation. The contractor shall furnish test data to prove that the roller is capable of operating in an ambient temperature of 120° F without exceeding the engine manufacturers recommended limits for radiator top tank temperature of applicable engine sump and gallery oil temperatures and hydraulic component manufacturers hydraulic fluid temperature limits.

4.5.2.3.1 Failure criteria. Failure of the contractor to provide test data or manufacturer's certification (see 3.12) shall constitute failure of this test.

4.5.2.4 Test procedures of lifting provisions. Test procedures of lifting provisions shall be as follows:

a. Lift the roller and hold in suspension in its normal travel position using slings that converge not more than 24 feet above the lowest extremity of the roller. Determine the forward angles of application for each provision. Measure the clearance between each sling and the roller. Measure eye openings and clearance dimensions of the provisions. Measure the load carried by each provision.

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b. Restrain the roller by anchoring the main frame and subject each provision to a force equal to the load factor specified in MIL-STD-209 times the load it carried in paragraph (a) above. An alternate method of test is to lift the roller as in (a) above and add weights to the main frame until the required force is obtained. Hold the roller under either of the above loading conditions for at least 90 seconds.

4.5.2.4.1 Failure criteria. Weld failure or permanent deformation caused by lifting or failure to meet the requirements of MIL-STD-209, shall constitute failure of this test.

4.5.2.5 Test procedures of tiedown provisions. With the roller anchored by means other than the provisions being tested, apply the proportion of the static loads as specified in MIL-STD-209 to each attachment. Measure dimensions of the tiedown.

4.5.2.5.1 Failure criteria. Failure to meet the requirements of MIL-STD-209 shall constitute failure of this test.

4.5.2.6 Electromagnetic interference. Unless otherwise specified, the roller shall be tested in accordance with SAE J551. A report tabulating the allowable maximum interference and the measured interference shall be furnished with the preproduction model test report.

4.5.2.6.1 Failure criteria. Failure to meet the requirements of SAE J551, shall constitute failure of this test.

4.5.2.7 Sound level test. Unless otherwise specified, the sound level tests shall be conducted (see 6.2 and 6.7). Frequency measurements, db(A) levels, tape recording, instrument identification, environmental information, and main sound source identification, shall be reported.

4.5.2.7.1 Failure criteria. One or more functional defects shall constitute failure, and acceptance of the equipment shall not be approved until all defects are corrected.

4.5.2.8 Quality conformance operational run-in. Each roller shall be serviced as required and shall be run-in in accordance with the manufacturer's normal procedures for a period of not less than 60 minutes.

4.5.2.8.1 Failure criteria. One or more functional defects shall constitute failure, and acceptance of the equipment shall not be approved until all defects are corrected.

4.6 Inspection comparison. The Government may select rollers at any time during the contract production period and subject these rollers to the examination specified in 4.5.1 and the tests specified in 4.5.2 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on units selected at random from those which have been accepted by the Government, and will not

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include the previously inspected preproduction model rollers. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.6.1 Inspection failure. Failure of an inspection comparison roller to meet any requirements specified herein during and as a result of the examination and test specified in 4.6 shall be cause for rejection of the inspection comparison roller(s) and shall be cause for refusal by the Government to continue acceptance of the production rollers until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on rollers previously accepted and produced under contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all rollers accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.7 Inspection of packaging. The preservation, packing, and marking shall be examined in accordance with the quality assurance provisions of MIL-R-3075.

5. PACKAGING

5.1 Preservation, packing and marking. Each complete roller shall be preserved, packed and marked in accordance with MIL-R-3075. The degree of preservation shall be level A, B, or commercial and the degree of packing shall be level A, B, or commercial, as specified (see 6.2).

6. NOTES

6.1 Intended use. The rollers are intended for compaction of earth fill, gravel, and soil stabilization areas. The rollers are intended to be air transportable within the range of an air transportable division.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Type of roller required (see 1.2).
c. Date of issue of DoDISS applicable and exceptions thereto (see 2.1.1).
d. Time frame for submission of first article roller and number of rollers required (see 3.2).
e. When the Government will conduct any or all of the first article examination and tests. When the Government will conduct some but not all of the first article examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests will be conducted by the contractor (see 3.2).

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f. When preproduction roller inspection is not required (see 3.2).
g. When draw bar shall be compatible with a crawler tractor drawbar conforming to SAE J249a pin size 1.75 (see 3.7.2).
h. When remote control system is required (see 3.12.5)
i. When electromagnetic interference requirements are other than as specified (see 3.17).
j. When fungus and moisture resistance is not required (see 3.19).
k. When the finishing coat shall be other than the lusterless forest green (see 3.21).
l. When sound measurement tests are not required.
m. Degree of preservation and degree of packing required (see 5.1).

6.3 Preproduction model. Any changes or deviations of production rollers from the approved preproduction of the model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish rollers conforming to this specification.

6.4 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor maintenance and operation manual to be furnished with each roller.

6.5 Provisions. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of rollers.

6.6 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.4).

6.7 Sound level. When the sound level of the equipment exceeds 85 dB(A) at the operator's station, a noise hazard caution sign per MIL-STD-1474 shall be permanently affixed to the equipment in the operator's compartment. The 85 dB(A) contour shall be defined. The noise hazard zone shall be clearly stenciled on the equipment in a conspicuous location with 3/4 inch letters stating: "Hearing protection required within 20 feet from this equipment".

Custodians:
Army - ME

Preparing activity:
Army - ME

User activity:
Army - CR

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